



Darwin visited the Galapagos Islands in 1835; the animals and plants he observed there are said to have contributed to his becoming an evolutionist, but it wasn't until 1859 that Darwin consolidated his observations into a book—*The Origin of Species*.

**ON NOVEMBER 24, 1859,** **CHARLES DARWIN** PUBLISHED his epoch-making book *On the Origin of Species*. In it he explained in detail his theory of evolution, first introduced in 1858. Darwin's theory was that species change and develop automatically through a **process of natural selection**; this idea was neatly summed up by the philosopher Herbert Spencer (1820–1903) as “**survival of the fittest**.” In his book, Darwin explained how, occasionally, a

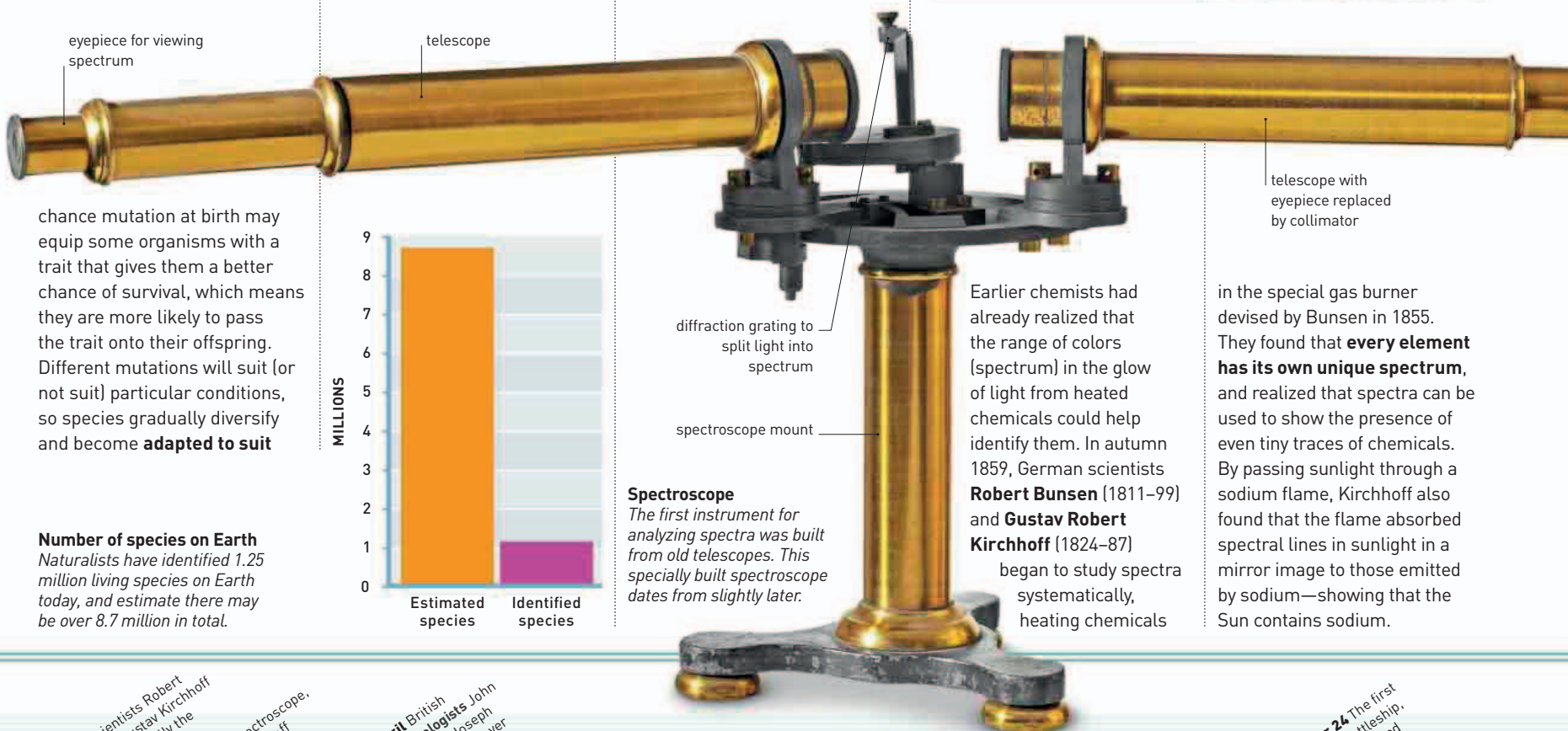
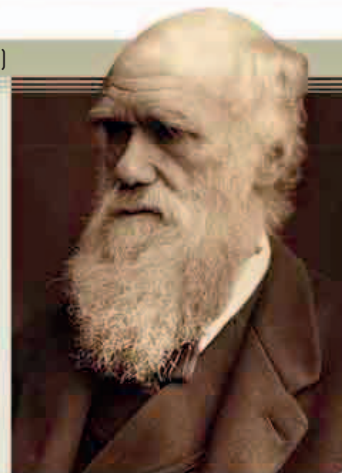
**specific habitats.** If habitats change, however, these special advantages may become weaknesses and as a result the species could die out.

What made Darwin's theory such a turning point was that the mechanism he proposed worked for all of life, and asserted that every organism is **descended from a common ancestor**. Although many people accepted the force of Darwin's arguments, some were bitterly critical and debate flared up.

In April, two British archaeologists, **John Evans** (1823–1908) and **Joseph Prestwich** (1812–96), made a startling discovery that **pushed human origins further back into prehistory**. In St. Acheul in northern France, they found a stone ax in layers that also contained fossils of extinct creatures, including mammoths. If mammoths and humans had existed together, then human life must date back tens of thousands of years.

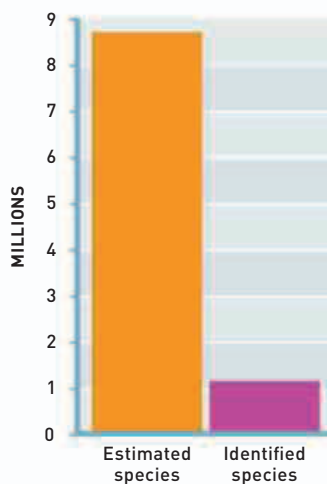
#### CHARLES DARWIN (1809–82)

Darwin trained as a doctor before embarking, as a naturalist, on a round-the-world voyage in 1831–36 aboard HMS *Beagle*. The voyage sowed the seeds for his theory of evolution by natural selection, which he finally revealed in 1858. He also applied his theory to humans in *The Descent of Man*, published in 1871.



chance mutation at birth may equip some organisms with a trait that gives them a better chance of survival, which means they are more likely to pass the trait onto their offspring. Different mutations will suit (or not suit) particular conditions, so species gradually diversify and become **adapted to suit**

**Number of species on Earth**  
Naturalists have identified 1.25 million living species on Earth today, and estimate there may be over 8.7 million in total.



**Spectroscope**  
The first instrument for analyzing spectra was built from old telescopes. This specially built spectroscope dates from slightly later.

Earlier chemists had already realized that the range of colors (spectrum) in the glow of light from heated chemicals could help identify them. In autumn 1859, German scientists **Robert Bunsen** (1811–99) and **Gustav Robert Kirchhoff** (1824–87) began to study spectra systematically, heating chemicals

in the special gas burner devised by Bunsen in 1855. They found that **every element has its own unique spectrum**, and realized that spectra can be used to show the presence of even tiny traces of chemicals. By passing sunlight through a sodium flame, Kirchhoff also found that the flame absorbed spectral lines in sunlight in a mirror image to those emitted by sodium—showing that the Sun contains sodium.

German scientists Robert Bunsen and Gustav Kirchhoff study systematically the glows of elements

Using a spectroscope, Gustav Kirchhoff discovers that the Sun contains sodium

April British archaeologists John Evans and Joseph Prestwich discover ancient stone tools

French engineer Etienne Lenoir produces the first two-stroke gasoline engine

Russian astronomer Marian Kowalski develops a way to work out the rotation of the Milky Way

August 28–September 2 The largest geomagnetic solar storm on record, Carrington Event

French mathematician Urbain Le Verrier proposes the existence of an undiscovered planet, Vulcan, to explain Mercury's orbit

November 24 Charles Darwin's *On the Origin of Species* is published

November 24 The first iron-hulled battleship, the *Gloire*, launched in France